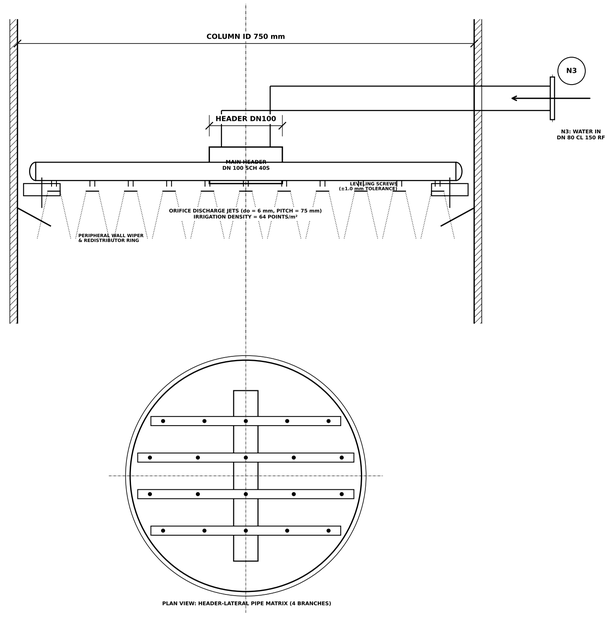
DETAIL A — HEADER-AND-LATERAL LIQUID DISTRIBUTOR

SECTIONAL ELEVATION & PLAN VIEW (SCALE 1:10 METRIC)



DETAIL B — WIRE MESH DEMISTER (MIST ELIMINATOR) ASSEMBLY

CROSS-SECTION & HOLD-DOWN DETAILS (SCALE 1:8 METRIC)

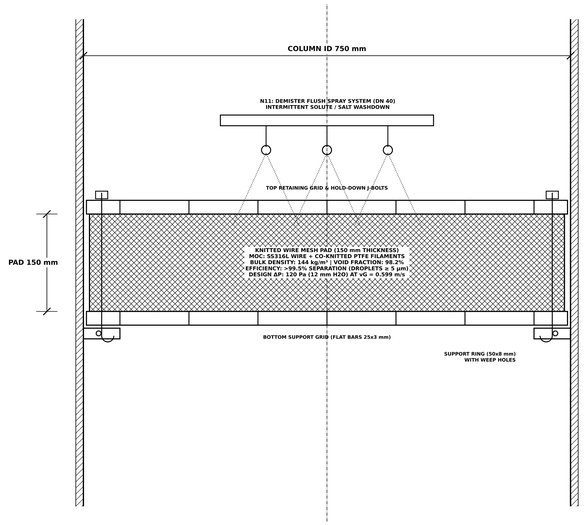
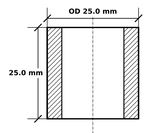


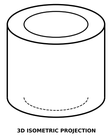
TABLE 1 — ABSORBER INTERNALS MECHANICAL DESIGN & HYDRAULIC SUMMARY

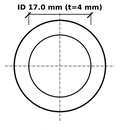
INTERNAL COMPONENT	DESIGN PARAMETERS / MOC	HYDRAULIC FUNCTION / CAPACITY
Liquid Distributor	Header-Lateral (4 branches) SS316L / PVDF lined	Irrigation density = 64 pts/m² Orifice velocity = 1.65 m/s
Liquid Redistributor	Perforated cone + wall wiper Titanium Gr 2 / PTFE	Collects wall flow (<5% wall channeling) Redistributes across active bed
Packing Hold-Down	Heavy-duty grid + mesh SS316L (50x5 mm frame)	Prevents ring lifting & bed expansion Design upward gas thrust = 1.8 kN
Raschig Ring Bed	**25-mm Ceramic Raschig Rings** **4500 mm Depth (Vp = 1.79 m³** H_OG = 0.84 m (Onda correlation)	Total gas transfer units N_OG = 5.09
Packing Support Grid	Multi-beam gas-injection Free flow area > 85%	Carries 16.1 kN normal operating load Prevents premature flooding at support
Gas Inlet Deflector	Inverted 60° conical chevron Titanium Gr 2 (4 mm thk)	Reduces gas inlet velocity from 14.1 m/s to uniform column velocity 0.599 m/s
Mist Eliminator	Knitted wire mesh (150 mm) SS316L + PTFE composite	>99.5% droplet removal (≥5 µm) Pressure drop ΔP = 120 Pa at design flow
Demister Spray Wash	Full-cone spray nozzles (3x) DN 40 manifold (SS316L)	Periodic flush of captured bromine Prevents mesh salt fouling and high ΔP
Anti-Vortex Baffle	Cross-plate baffle (4 mm thk) SA-516 Gr 70 + PVDF lined	Suppresses liquid vortexing & air entrainment Ensures steady suction to bottoms pump

DETAIL C — 25-mm CERAMIC RASCHIG RING & BED DETAIL

3D ISOMETRIC, ORTHOGRAPHIC CAD & PROPERTIES (SCALE 1:1 METRIC)







RASCHIG RING DESIGN DATA

Nominal Ring Size:	25 mm (1.0 inch)
Material of Constr.:	Glazed Chemical Stoneware
Standard / Code:	ASTM C515 Class 1
Specific Surface (ap):	**190 m²/m³**
Bed Void Fraction (ε):	0.74 (74.0% Void)
Bulk Density (ρbulk):	680 kg/m³
Packing Factor (Fp):	100 m⁻¹ (30.48 ft²/ft³)
Acid Resistance:	>99.8% (Boiling H2SO4)
Water Absorption:	<0.5% (Non-porous)
Compressive Strength:	>150 MPa
Active Bed Height (Z):	**4500 mm (Calculated)**
Active Bed Volume (Vp):	1.79 m³ (Selected 1.99 m³)
Dry Packing Mass (Mp):	1220 kg (1.22 tonnes)
Dry Gravity Load (Wp):	**11.9 kN (Calculated)**
Flooded Holdup Load:	12.9 kN (Contingency)
Support Design Load:	16.1 kN (26.4 kN Upset)

THIRD ANGLE PROJECTION

DETAIL D — BOTTOM GAS INLET & AERODYNAMIC DEFLECTOR

CROSS-SECTION & FLOW DISTRIBUTION INTERNALS (SCALE 1:8 METRIC)

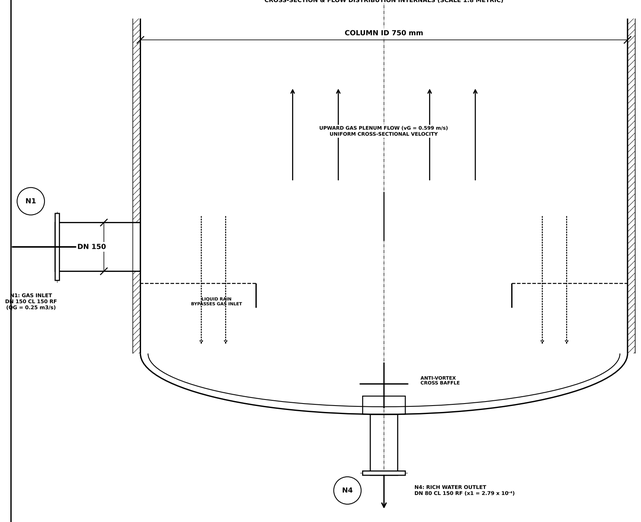


TABLE 2 — BILL OF MATERIALS (BOM) FOR ABSORBER INTERNALS

ITEM	DESCRIPTION	QTY	MATERIAL SPECIFICATION	WEIGHT (KG)
1	Mist Eliminator Pad	1 SET	SS316L Wire + PTFE Knitted Mesh	18.5 kg
2	Demister Retaining Grids	2 PCS	SS316L Flat Bars (25x3 mm)	24.0 kg
3	Demister Flush Header	1 SET	DN 40 Sch 40S SS316L + 3 Nozzles	12.2 kg
4	Liquid Distributor Manifold	1 SET	DN 100/50 Sch 40S + PVDF Liner	45.0 kg
5	Wall Wiper Rings	2 SET	Titanium Gr 2 (3 mm Sheet)	14.8 kg
6	Hold-Down Top Grid	1 SET	SS316L Heavy Mesh + Ring Frame	28.5 kg
7	Ceramic Raschig Rings	1.79 m³	ASTM C515 Glazed Ceramic (25 mm)	1220.0 kg
8	Gas-Injection Support Grid	1 SET	High-Strength FRP / Titanium Gr 2	65.0 kg
9	Support Ring & Gussets	1 SET	SA-516 Gr 70 + 3 mm PVDF Clad	42.0 kg
10	Gas Inlet Deflector Cone	1 PC	Titanium Gr 2 / PVDF Plate (4 mm)	16.5 kg
11	Sump Anti-Vortex Baffle	1 SET	SA-516 Gr 70 + PVDF (4 mm Pit)	11.0 kg

INTERNALS FABRICATION, TOLERANCES & INSTALLATION NOTES

1. DISTRIBUTOR HORIZONTAL LEVELING TOLERANCE:

- DISTRIBUTOR LATERALS SHALL BE LEVELED WITHIN +/- 1.0 mm ACROSS THE FULL 750 mm COLUMN DIAMETER.

- VERIFICATION TO BE CONDUCTED USING WATER FLOODING TEST PRIOR TO PACKING CHARGING.

2. PACKING BED CHARGING & FLOTATION PREVENTION:

- 25-mm CERAMIC RASCHIG RINGS SHALL BE DUMPED INTO A WATER-FILLED COLUMN (WET PACKING METHOD)

TO CUSHION RING IMPACT AND PREVENT CHIPPING / BREAKAGE OF PORCELAIN WALLS.

- TOP HOLD-DOWN GRID MUST BE FASTENED SECURELY TO SHELL CLEATS TO PREVENT RING FLUIDIZATION.

3. SUPPORT GRID FREE AREA REQUIREMENT:

- SUPPORT GRID FREE FLOW AREA MUST EXCEED 85% OF COLUMN CROSS-SECTIONAL AREA TO ELIMINATE

BOTTLENECKING OF GAS INJECTION AND LOCAL PREMATURE FLOODING AT THE BED BASE.

4. CORROSION & MATERIAL SELECTION RIGOR:

- DUE TO EXTREME AGGRESSIVENESS OF WET BROMINE (Br2 + H2O), ALL METALLIC FASTENERS EXPOSED

TO PROCESS VAPORS MUST BE HASTELLOY C-276 OR TITANIUM GRADE 2 (ASTM B348 / F468).

- GASKETS FOR INTERNAL FLANGES: VIRGIN PTFE ENVELOPE GASKETS (100% CORROSION RESISTANT).

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS EQUIPMENT DESIGN & DRAWING

DRAWING TITLE PACKED-BED ABSORBER (C-101) — INTERNAL ASSEMBLY & PROCESS INTERNALS DETAILED CAD	**LAB / SUBJECT** PROCESS EQUIPMENT DESIGN & DRAWING	**COURSE / BRANCH** B.E. CHEMICAL ENGINEERING	**SCALE & REVISION** SCALE: DETAILED VIEWS 1:1, 1:8, 1:10 METRIC | REV: 00
STUDENT NAME SOURISH SENAPATI	**EXAM ROLL NUMBER** 002310301035	**DRAWING / DOCUMENT NO.** PEDD-PBA-02 (SHEET 02 OF 05)	**DATE** 29 / 09 / 2026
EQUIPMENT IDENTIFICATION & SERVICE ABSORBER PROCESS INTERNALS — DISTRIBUTOR, DEMISTER, PACKING & DEFLECTOR	**DESIGN BASIS & METALLURGY / LINING** ASME SEC VIII DIV 1 / SA-516 GR 70 + PVDF LINER / SS316L / CERAMIC	**PROCESS PERFORMANCE & RECOVERY** 25-mm CERAMIC RASCHIG RINGS | AP = 190 m²/m³ | ε = 0.74 | L = 17.25 kg/s	**DESIGN STATUS CLASSIFICATION** CALCULATED: ID 750, Z 4500 | PRELIMINARY CAD: SHELL & NOZZLES

DIMENSIONS: METRIC (MM / M) | FLANGES: ASME B16.5 CLASS 150 RF | WELDING: ASME SEC IX | THIRD ANGLE PROJECTION